MSP-O: A Two-Stage Mask Pretraining Method With Object Focus for Remote Sensing Object Detection

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***Abstract*— Pretraining is showing fundamental importance in computer vision, to initialize the backbones of deep learning models with large natural image datasets, e.g., ImageNet. However, there is seldom a special pretraining technique focused on remote sensing object detection due to the distinct characteristics of remote sensing images, such as specific imaging angles, complex backgrounds, and various target categories. This article aims to bridge the gap between natural image data and remote sensing image data and to facilitate pretraining for the specific object detection task. This article introduces masked remote sensing pretraining with object (MSP-O), an innovative two-stage masked pretraining method. Traditional single-stage pretraining frameworks aim to extract intrinsic features from large-scale image datasets to learn object discrimination and localization ability. MSP-O builds upon this by adding a second-stage pretraining phase to learn specific features related to remote sensing objects of interest themselves, facilitating the downstream object detection task. Specifically, MSP-O incorporates a pixel-level restoration task focused on the objects of interest, guiding the model to prioritize the objects to be detected. Compared with traditional pretraining methods, MSP-O is able to learn feature representations that are more relevant to the detection task, especially when the data scale is limited. Extensive experiments on the DOTA and DIOR datasets demonstrate the effectiveness of MSP-O, showing substantial improvements in detection performance compared with conventional methods.**

***Index Terms*— Deep learning, object detection, remote sensing, two-stage pretraining.**

I. Introduction

PRETRAINING on large-scale datasets is a preliminary and essential step in developing powerful deep learning models [1]. Recent years have seen the success of pretraining in natural language processing [2] and computer vision [3], [4], [5], [6]. Pretraining can be mainly categorized into supervised learning and self-supervised learning (SSL). Supervised learning depends on abundant annotated data for specific-task model training. For instance, ResNet [3] and Vision Transformer (ViT) [7] excel in downstream tasks using ImageNet pretrained weights [8]. In remote sensing, Long et al. [9] used the million-scale MillionAID dataset [10] to validate remote sensing visual foundation models across multiple tasks such as scene recognition, semantic segmentation, object detection, and change detection with pretrained convolutional neural network (CNN) and ViT backbone weights [7]. Findings show that supervised pretraining in remote sensing usually boosts performance when downstream and upstream task objectives align.

In contrast, SSL acquires discriminative feature representations by extracting intrinsic data patterns, demonstrating three critical advantages: reduced annotation dependency, enhanced generalization capacity, and optimized data efficiency. This framework employs pseudolabeled unlabeled data for large-scale pretraining to learn domain-agnostic visual primitives, enabling effective capture of task-specific discriminative features even with limited supervision. Recent efforts have developed comprehensive remote sensing datasets spanning multitemporal/spatial scales for SSL pretraining [11], [12], with demonstrated success in semantic segmentation through invariant feature learning [13].

Masked image modeling (MIM), a leading SSL approach first introduced by He et al. [14], utilizes a ViT-based encoder–decoder architecture to reconstruct randomly masked image patches, effectively learning robust and generalizable visual representations. The success of MIM-based pretraining has been extensively validated in remote sensing applications [15], [16], [17], [18], [19]. The MIM paradigm learns augmentation-invariant representations through image restoration objectives, achieving state-of-the-art performance in pixelwise prediction tasks. This capability proves particularly vital for cross-condition feature preservation in remote sensing analysis, where target objects exhibit stable spectral–spatial signatures across heterogeneous acquisition scenarios (varying illumination, sensor parameters, or seasonal changes) [14], [20], [21]. Wang et al. [22] found MAE beats other pretraining methods, especially on small, uneven datasets. This highlights the need for adequate ViT pretraining [7]. Systematic ablation studies by Long et al. [9] further confirmed MIM's ability to enhance model generalization while reducing computational demands in large-scale visual processing tasks. Recent methodological advancements have specifically addressed remote sensing challenges. Wang et al. [16] proposed rotated varied-size window attention to handle large arbitrarily oriented objects. RingMo framework [15] imple-

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mented random masking strategies to preserve small object information, albeit with inherent limitations in size adaptation due to stochastic masking patterns. Singh et al. [23] demonstrated the critical role of MAE-based pretraining for model initialization in large-scale remote sensing applications. Geospatial foundation models [25] use the multiobjective continual pretraining paradigm, which is initialized with ImageNet-22k pretrained weights, and the systematic integration of contextual constraints within reconstruction loss functions proposed by context-enhanced MIM [25] to improve feature representation learning. Notably, directly applying MIM for Transformer-based pretraining [26] in remote sensing demands domain-specific adaptations. This requirement stems from the limited scale of aerial object detection datasets compared with natural image corpora [27]. Such data scarcity risks inducing pretraining overfitting, which can propagate error signals to downstream tasks during fine-tuning [27].

However, pretraining techniques specific for remote sensing object detection remain limited [1]. These existing techniques not only have limitations in themselves but also pose practical challenges in downstream remote sensing tasks. For instance, the wide variety of target scales and the long-tail data distribution are common issues. Therefore, to improve the generalization ability of models and the accuracy of detection, it is imperative to develop more specialized pretraining strategies.

The principal challenge stems from the inherent domain shift between pretraining and downstream fine-tuning distributions, exacerbated by a pervasive class imbalance in object detection tasks. While large-scale pretraining datasets (e.g., Million-AID [10]) offer rich feature learning opportunities through abundant samples, their efficacy diminishes when downstream task characteristics substantially diverge from the pretraining domain. For instance, critical scarcity in underrepresented categories (e.g., aircraft and bridges) induces suboptimal task adaptation during fine-tuning, manifested through degraded feature transferability. Furthermore, the long-tailed class distributions, such as those prevalent in benchmarks like DIOR [28] and DOTA [29], significantly impair model discriminative capacity, as shown in Fig. 1. There are three issues: inadequate feature separability for sparse classes, gradient update bias toward dominant categories, and suboptimal decision boundary formation. Dataset-aware architectures should align pretraining with downstream tasks through three strategies: cross-domain feature compatibility analysis, adaptive reweighting for class imbalance, and domain-invariant representation learning integration. This approach ensures knowledge transfer while maintaining discriminative capabilities via balanced feature uniformity and controlled embeddings.

Another critical challenge stems from the intrinsic task mismatch between self-supervised pretraining paradigms and supervised detection fine-tuning objectives. Conventional pretraining strategies predominantly focus on either holistic structural comprehension through whole-image reconstruction (emphasizing low-frequency spatial patterns) or localized discriminative modeling via contrastive learning (prioritizing high-frequency texture features) [30]. Conversely, object detection frameworks demand simultaneous requirements for multi-instance semantic categorization and geometrically precise bounding box regression. This fundamental divergence creates representation transfer bottlenecks, particularly evident in compromised feature utilization within region proposal networks and classifier modules [31], [32]. The tension arises from pretrained features being optimized for either global scene understanding or local texture discrimination, neither fully addressing the dual demands of instance-level recognition and pixel-accurate localization required in detection pipelines.

Natural images are predominantly characterized by variations in the shapes of objects and the viewing perspectives. In contrast, remote sensing images possess distinct and consistent features. The repetitive structural patterns that are inherently present in urban elements, such as buildings, roads, and transportation hubs, are particularly noteworthy. These images generally maintain stable spatial patterns across a wide range of viewing angles. This inherent stability enables models to generalize effectively from limited training data to novel environments. Consequently, models that can learn discriminative target attributes from small-scale datasets can successfully transfer this knowledge to similar targets in diverse scenarios, thereby enhancing both their generalization ability and overall detection performance.

To address these domain-specific challenges in remote sensing object detection while preserving the unique characteristics of aerial imagery, this article proposes a masked remote sensing pretraining with objects (MSP-O) framework a novel two-stage paradigm that bridges the gap between natural image-based pretraining and detector-specific fine-tuning. Unlike conventional approaches that directly apply natural image priors or train detectors from scratch, MSP-O establishes hierarchical feature learning through the following.

1) *Generalized-to-Specific Representation Transfer:* The first phase preserves universal feature encoding capabilities through structural pretraining on large-scale natural image datasets, while progressively adapting to downstream task patterns in aerial imagery without discarding domain-agnostic visual priors.
2) *Architecture-Adaptive Refinement:* The second phase introduces a pluggable enhancement module compatible with mainstream Transformer architectures, enabling seamless integration with downstream detectors through parameter-efficient adaptation.
3) *Data-Efficient Learning:* Our proposed copy-embedding strategy addresses small-scale dataset limitations by reusing latent features from the pretraining phase, requiring no additional annotations beyond existing bounding box labels.

The proposed approach enhances conventional pretraining paradigms by introducing an intermediate pretraining phase between generic feature extraction and task-specific fine-tuning. This transitional stage reinforces domain-invariant representations through object-aware masked reconstruction that preserves geometric structures. Comprehensive experiments on DOTA and DIOR benchmarks demonstrate the method's effectiveness in both intradataset evaluation and cross-scenario transferability.

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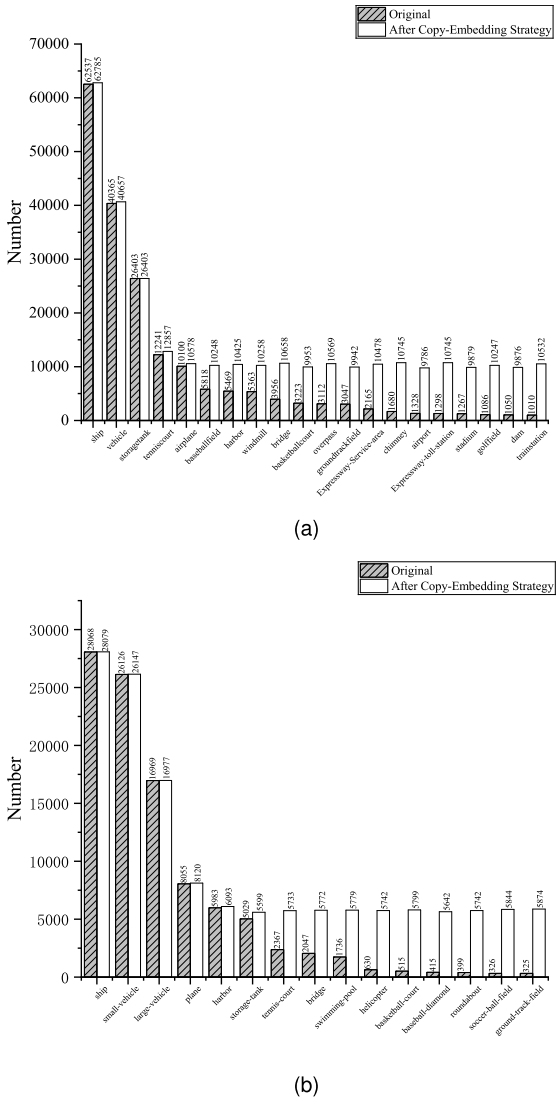


Fig. 1. Distribution of the number of categories before and after copy-embedding strategy regarding (a) DIOR and (b) DOTA datasets.

II. Proposed Method

A. Copy-Embedding Strategy

Some remote sensing datasets often have a significant class imbalance. In Fig. 1, the DIOR comprises a huge number of stations and vehicles but a small number of ships and airplanes. On the other hand, DOTA consists of many images related to ships but few images of stadiums and stations. This imbalance poses challenges for object recognition in remote sensing images.

To address the challenge of data imbalance and improve pretraining efficiency, we propose a novel copy-embedding strategy, as illustrated in Fig. 2. This approach comprises two key phases: image selection and category balancing.

1) Phase 1 Image Selection: In the data selection phase, we start with the original datasets, such as DOTA, and utilize a perceptual hashing algorithm (PHA) to remove redundant and highly similar images [33]. This is depicted in the transition from Fig. 2(a) to (c), where Fig. 2(c) represents the dataset after PHA application. Next, we employ the ViT backbone network to compute the perceptual loss, further refining the dataset by eliminating images with overlapping features, as shown in Fig. 2(d), which represents the final dataset. This process reduces dataset size while preserving diversity, thereby mitigating the risk of overfitting redundant data. The proposed hierarchical filtering pipeline formalizes the data selection process through function composition. Given a raw dataset $\mathcal{D}_{\text{raw}} \subset \mathcal{X}$, where $\mathcal{X} = \mathbb{R}^{H\times W\times 3}$ denotes the image space, we sequentially apply two complementary filtering operators

$$\mathcal{D}_{\text{filtered}} = \Phi_{\text{ViT}} \circ \Phi_{\text{PHA}}(\mathcal{D}_{\text{raw}}) \tag{1}$$

where $\Phi_{\text{PHA}} : \mathcal{X} \to \mathcal{X}$ denotes the perceptual hashing filtering [33]

$$\Phi_{\text{PHA}}(x_i) = \begin{cases} x_i, & \text{if } \min\limits_{j<i} \|h(x_i) - h(x_j)\|_0 > \tau \\ \emptyset, & \text{otherwise} \end{cases} \tag{2}$$

with $h(\cdot) : \mathbb{R}^{H\times W\times 3} \to \{0,1\}^{256}$ being the perceptual hash function and $\tau \in \mathbb{Z}^+$ the Hamming distance threshold. Subsequent ViT-based filtering $\Phi_{\text{ViT}} : \mathcal{X} \to \mathcal{X}$ removes feature-space redundancies

$$\Phi_{\text{ViT}}(x_i) = \begin{cases} x_i, & \text{if } \max\limits_{x_j\in\mathcal{D}_{\text{PHA}}} \dfrac{f(x_i)^\top f(x_j)}{\|f(x_i)\|\|f(x_j)\|} < \eta \\ \emptyset, & \text{otherwise} \end{cases} \tag{3}$$

where $f(\cdot) : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^d$ represents ViT feature embeddings and $\eta \in [0,1]$ the cosine similarity threshold.

2) Phase 2 Category Balancing: For the challenge of sample imbalance, we concentrate on enhancing the representation of underrepresented categories. We achieve this by selecting and augmenting samples within bounding boxes using the techniques detailed in Table I. The images of the selected datasets are depicted in Fig. 2(f), while Fig. 2(g) exemplifies the augmentation of minor objects. These objects are then subjected to the transformations listed in Table I to produce Fig. 2(h). The augmented samples are strategically embedded into random positions within other images with collision avoidance, as shown in Fig. 2(i). This approach not only increases the sample size for underrepresented classes through controlled replication but also optimizes computational efficiency by constraining dataset expansion. Crucially, it preserves an optimal balance between sample quantity and feature diversity, thereby enhancing model generalization across unseen scenarios under limited data conditions. The class distribution comparison before and after applying the

TABLE I
Data Augmentation Methods Used by Copy Embedding

Transformation Method	Parameters
Random Rotation	$0 \sim 90°$
Random Flipping	x, y, c
Random Scaling	$0.5 \sim 1.5\times$
Random Stretching	$0.5 \sim 1.5\times$
Random Affine	$0 \sim 90°$
Random Color to Gray	0.2 (probability)

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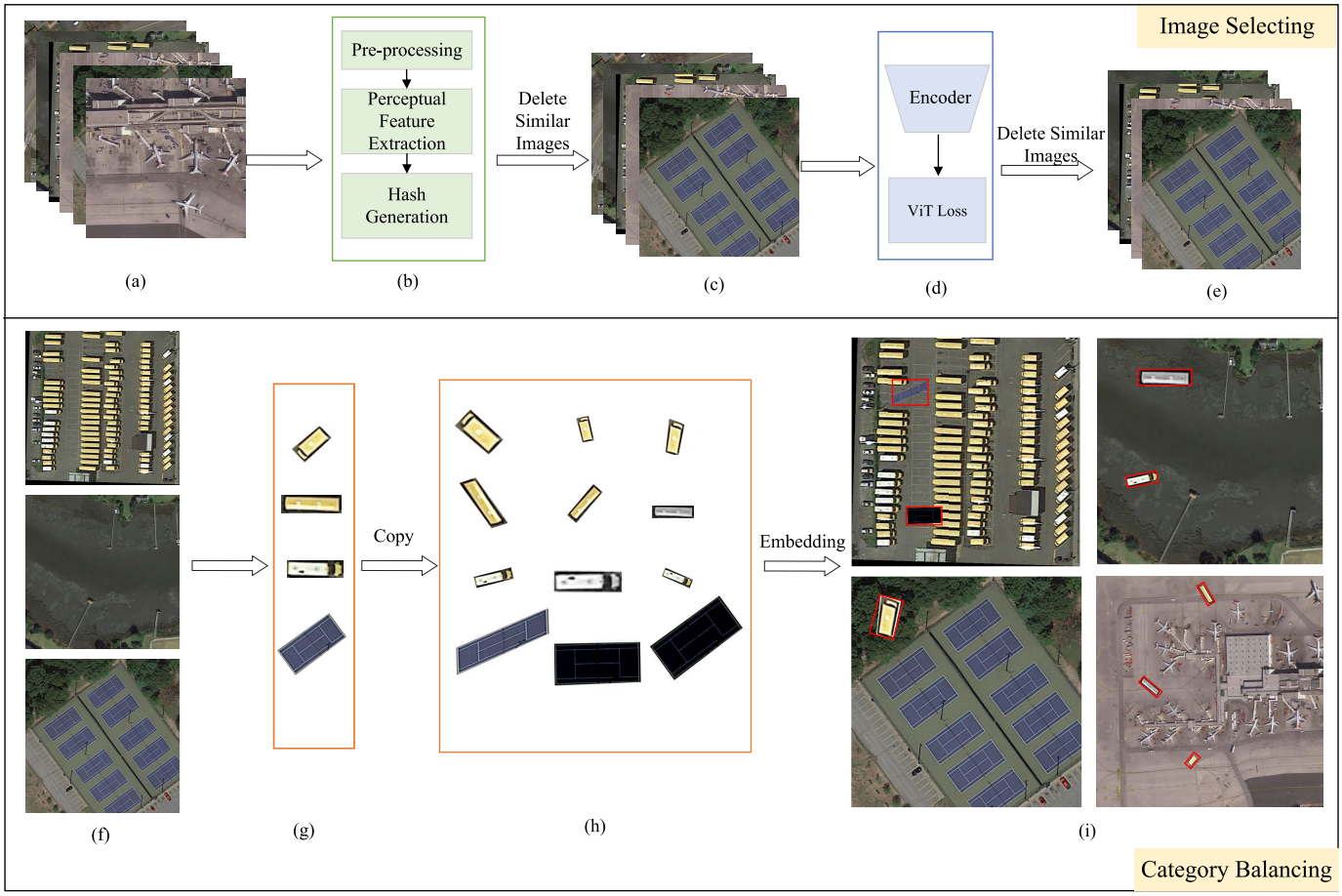


Fig. 2. Flowchart of copy-embedding strategy, which consists of two parts: image selection and category balancing. The red box represents the object for embedding; it is solely for visualization purposes and does not appear in the dataset. (a) Original datasets. (b) PHA. (c) After PHA. (d) ViT perceptual. (e) Final datasets. (f) Images. (g) Sample. (h) Augmentation. (i) Final images.

copy-embedding strategy is quantitatively demonstrated in Fig. 1.

For target minority class $c \in \mathcal{C}$ with instance count N_c, we synthesize new samples through

$$\hat{\mathcal{D}}_c = \bigcup_{k=1}^{K} \left[\mathcal{T}_k(\mathcal{B}_c) \boxplus \mathcal{P}_{\text{non-overlap}} \right] \tag{4}$$

where $\mathcal{B}_c \subset \mathcal{D}_{\text{filtered}}$ denotes original bounding boxes for class c, $\mathcal{T}_k : \mathbb{R}^5 \to \mathbb{R}^5$ denotes the geometric transformations in Table I, and $\boxplus : \mathbb{R}^5 \times \mathbb{N}^2 \to \mathbb{R}^5$ denotes the collision-aware spatial embedding operator. The position selection are as follows:

$$\mathcal{P}_{\text{non-overlap}} = \left\{ p \in \mathcal{P}_{\text{rand}} \,\middle|\, \max_{b_j \in \mathcal{B}_{\text{base}}} \text{IoU}(p, b_j) < \tau \right\} \tag{5}$$

where $\mathcal{B}_{\text{base}}$ represents existing bounding boxes in the background image, $\text{IoU} : \mathbb{R}^4 \times \mathbb{R}^4 \to [0, 1]$ computes intersection-over-union, and $\tau = 0.1$ is the overlap threshold. The augmentation factor $K \in \mathbb{N}$ follows:

$$K = \left\lfloor \frac{N_{\text{majority}}}{\max(1, N_c)} \right\rfloor_{\text{clip}} \tag{6}$$

with $\lfloor \cdot \rfloor_{\text{clip}} : \mathbb{R} \to \mathbb{N}$ enforcing $K \leq K_{\max}$ to prevent over-augmentation.

The final composite image generation combines background $x_b \in \mathbb{R}^{H \times W \times 3}$ and augmented instances $x_a \in \mathbb{R}^{h \times w \times 3}$ through

$$x_{\text{final}} = \Psi_{\text{blend}}(x_b, M \odot x_a) \tag{7}$$

where $M \in \{0, 1\}^{H \times W}$ denotes the instance mask and $\Psi_{\text{blend}} : \mathbb{R}^{H \times W \times 3} \times \mathbb{R}^{H \times W \times 3} \to \mathbb{R}^{H \times W \times 3}$ the Poisson-aware blending operator.

The efficacy of copy-embedding is quantitatively validated in Fig. 1, demonstrating improved class distribution alignment while maintaining intersample diversity metrics within 5% of original values.

B. Mask Over Object Strategy

This article proposes a two-stage pretraining framework for remote sensing object detection, as illustrated in Fig. 3. The architecture introduces MSP-O, as shown in Fig. 3(b), which establishes an intermediate learning phase that bridges the initial pretraining stage, as shown in Fig. 3(a), and downstream fine-tuning, as shown in Fig. 3(c).

The first stage, depicted in Fig. 3(a), systematically applies multiple pretraining methodologies including classification-based learning, contrastive representation learning, and MIM. For implementation consistency, we initialize this stage using

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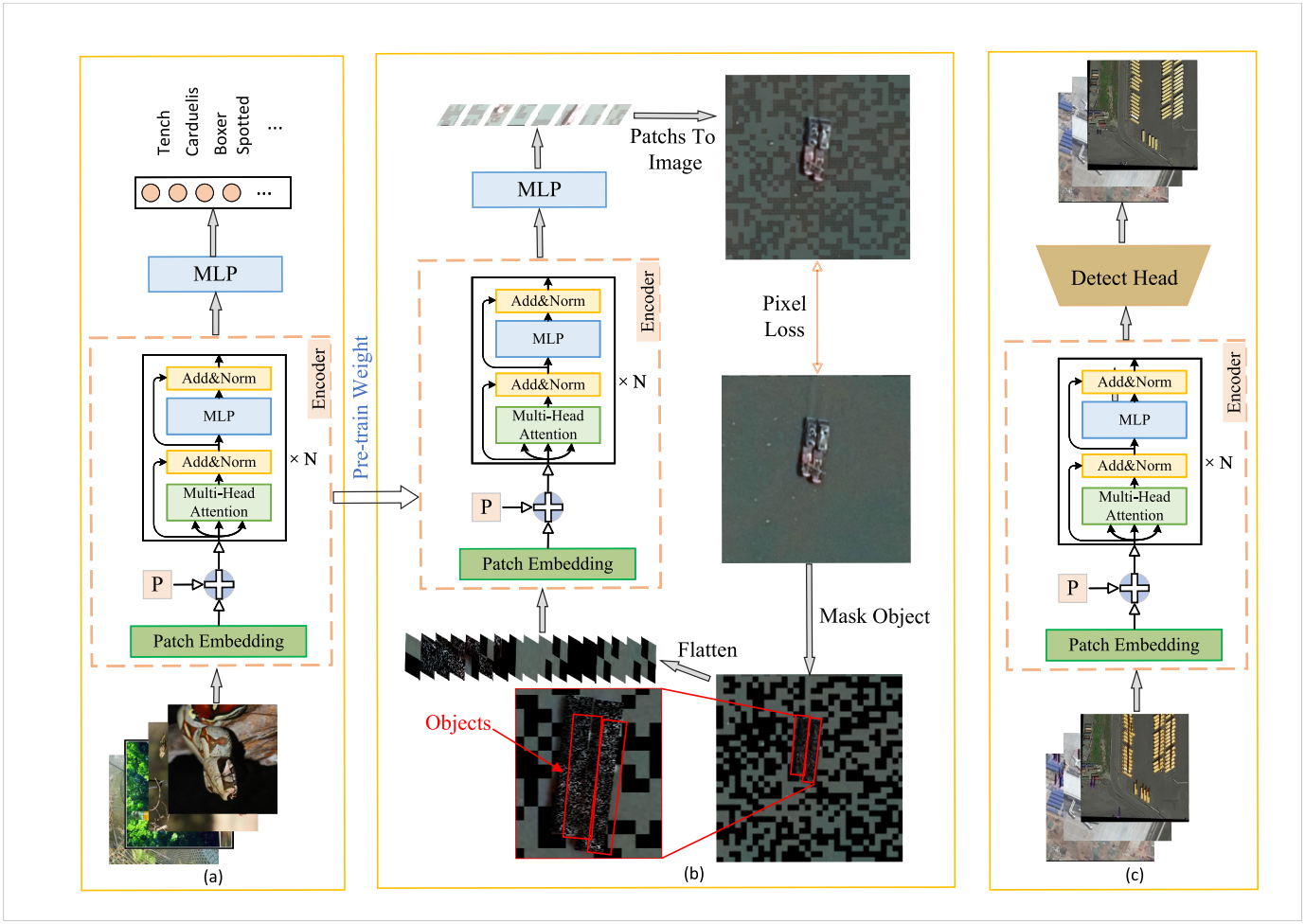


Fig. 3. Illustration of our method for training remote sensing object model. P stands for positional encoding. (a) First-stage pretraining. (b) Second-stage pretraining. (c) Object detector fine-tuning.

MAE pretraining weights following the methodology proposed by Wang et al. [16].

The dataset is formally defined as follows:

$$\mathcal{D} = \left\{ \left(I_k, \left\{ \ell_k^{(j)} \right\}_{j=1}^{m} \right) \right\}_{k=1}^{n} \subseteq \mathcal{I} \times \mathcal{L}^m.$$

1) *Symbol definitions:*
   a) $n \in \mathbb{N}^+$: Total number of images in the dataset.
   b) $m \in \mathbb{N}^+$: Fixed number of labels per image.
   c) $k \in \{1, \ldots, n\}$: Index of images.
   d) $j \in \{1, \ldots, m\}$: Index of labels for a specific image.
2) *Image space $\mathcal{I}$:*
   a) Each image $I_k \in \mathcal{I}$ is a 3-D tensor $\mathbb{R}^{H \times W \times 3}$.
   b) H: Height in pixels, W: Width in pixels, and 3: RGB channels.
3) *Label space $\mathcal{L}$:*
   a) Each label $\ell_k^{(j)} \in \mathcal{L}$ represents a rotated bounding box

$$\ell_k^{(j)} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix} \in \mathbb{R}^{4\times 2}.$$

1) Adaptive Masking Mechanism: The MSP-O strategy implements two-stage masking

$$\texttt{Mask}(I) = \Gamma_{\text{obj}}(I) \oplus \Gamma_{\text{bg}}(I). \tag{8}$$

1) *Background Masking:* Global masking with parameters $(p_{\text{bg}}, \rho_{\text{bg}})$

$$\Gamma_{\text{bg}}(I) = I \odot M_{\text{bg}}, \quad M_{\text{bg}} \sim \mathcal{G}(p_{\text{bg}}, \rho_{\text{bg}}). \tag{9}$$

2) *Object Masking:* Localized masking per bounding box b_j with $(p_{\text{obj}}, \rho_{\text{obj}})$

$$\Gamma_{\text{obj}}(I) = \bigcup_{j=1}^{m} \left[\mathcal{P}(b_j) \odot M_{\text{obj}}^{(j)} \right], \quad M_{\text{obj}}^{(j)} \sim \mathcal{G}(p_{\text{obj}}, \rho_{\text{obj}}). \tag{10}$$

1) $\mathcal{G}(p, \rho)$: Mask generator with patch size p and ratio ρ.
2) $\mathcal{P}(b_j)$: Box-to-mask projector $\mathcal{B} \to \{0, 1\}^{H \times W}$.
3) $\oplus$: Priority composition (objects override background).

2) Mask Fusion Operator: The fusion logic is explicitly defined as follows:

$$(A \oplus B)_{(h,w)} = \begin{cases} A_{(h,w)}, & \text{if } \sum_{j=1}^{m} \mathcal{P}(b_j)_{(h,w)} \geq 1 \\ B_{(h,w)}, & \text{otherwise.} \end{cases} \tag{11}$$

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3) Parameter Configuration: Hyperparameters are set as follows.

1) *Object:* $p_{\text{obj}} = 4$ and $\rho_{\text{obj}} = 0.75$ (high-density local masking).
2) *Background:* $p_{\text{bg}} = 16$ and $\rho_{\text{bg}} = 0.5$ (sparse global masking).

As visualized in Fig. 3(b), the masked image $\texttt{Mask}(I)$ is reconstructed through an autoencoder f_θ with

$$\hat{I} = f_\theta(\texttt{Mask}(I)) \in \mathcal{I} \tag{12}$$

where the encoder–decoder architecture f_θ and quantitative evaluation uses standard metrics [peak signal-to-noise ratio (PSNR)] as reported in Section III.

The final adaptation phase Fig. 3(c) involves detector-specific fine-tuning using the optimized weights from the MSP-O stage. This stage demonstrates compatibility with various established detection architectures, including but not limited to the oriented R-CNN framework presented in [34].

C. Design Principles of MSP-O Model

The MSP-O architecture adopts a dual-component framework comprising an encoder-decoder structure, extending the foundational paradigm established in SimMIM [20]. As depicted in Fig. 3(b), this design facilitates hierarchical feature learning through coordinated processing of visual information. The encoder module employs any standard Transformer-based visual backbone [26], including but not limited to ViT [7], ViTAE [35], or Swin Transformer [36], which share the common capability of processing image patches through self-attention mechanisms. Notably, our implementation preserves all input patches throughout the encoding phase, regardless of masking status, to maintain spatial integrity through learned positional embeddings. The encoding process initiates with image partitioning into fixed-size patches, each converted to embedding vectors through linear projection. Spatial relationships are explicitly encoded via additive positional embeddings before the sequence enters multiple Transformer blocks. Each block sequentially applies self-attention operations and feed-forward transformations, progressively building hierarchical feature representations that capture both local details and global contextual relationships. For computational efficiency, the decoder adopts a minimalist design with only a single linear projection layer. This architectural simplification achieves dual objectives: maintaining the encoder-derived high-dimensional feature relationships while simultaneously reducing parametric complexity to facilitate direct masked patch reconstruction. The resulting computationally efficient architecture demonstrates particular operational advantages in resource-limited implementations, achieving optimal feature utilization efficiency without compromising reconstruction capability through unnecessary structural redundancy.

D. Encoder Architecture

The encoder component employs a Transformer-based backbone [26] that supports various implementations and include the following.

1) Vision Transformer [7].
2) ViTAE with enhanced inductive bias [35].
3) Swin Transformer with shifted windows [36].

Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, the processing pipeline initiates with patch partitioning

$$\mathbf{P} = \phi(\mathbf{I}) \in \mathbb{R}^{N\times D} \tag{13}$$

where $\phi(\cdot)$ denotes the patch embedding operation, producing N patches with dimension D. Positional encoding $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{N\times D}$ is subsequently added to preserve spatial relationships

$$\mathbf{X}^{(0)} = \mathbf{P} + \mathbf{E}_{\text{pos}}. \tag{14}$$

The encoded features undergo L Transformer blocks containing multihead self-attention (MSA) and multilayer perceptron (MLP) operations

$$\mathbf{X}^{(l)} = \text{MLP}\big(\text{MSA}\big(\mathbf{X}^{(l-1)}\big)\big) + \mathbf{X}^{(l-1)}, \quad l = 1, \ldots, L. \tag{15}$$

Notably, the encoder processes all image patches in their original configuration regardless of masking status, maintaining complete spatial context throughout the transformation hierarchy.

E. Decoder Design

The decoder module adopts a parameter-efficient architecture formulated as follows:

$$\hat{\mathbf{P}} = \mathbf{W}_d\mathbf{X}^{(L)} + \mathbf{b}_d \tag{16}$$

where $\mathbf{W}_d \in \mathbb{R}^{D\times D}$ denotes the weight matrix and $\mathbf{b}_d \in \mathbb{R}^D$ the bias vector. This simplified design embodies three fundamental principles.

1) *Structural Minimalism:* Implements single-layer linear projection instead of multistage transformations.
2) *Dimensional Preservation:* Maintains feature space consistency ($\mathbb{R}^{N\times D} \to \mathbb{R}^{N\times D}$).
3) *Operation Linearity:* Eliminates nonlinear activation functions to prevent information distortion.

The architectural simplification is driven by two observations: deep encoders capture multiscale patterns, and pixel-level reconstruction uses linear transformations of rich features.

F. Loss Fuction

To guide the model in reconstructing high-quality remote sensing images rich in detail while focusing on structural accuracy and noise resilience, the loss function of the MSP-O framework comprises two components

$$L = L_{\text{Reconstruction}} + \alpha L_{\text{TV}} \tag{17}$$

where $L_{\text{Reconstruction}}$ computes pixel-level fidelity in masked regions using smooth $L1$ loss, chosen for its robustness to outlier gradients compared with the $L2$ norm. To prevent small objects from being overshadowed by larger background areas, we introduce an area-proportional weighting scheme

$$w = \mathbb{1} \circ \log\left(\frac{s_{\text{object}}}{s}\right). \tag{18}$$

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Algorithm 1 MSP-O Pretraining Algorithm

Require: Training dataset $\{(x_i, \mathcal{B}_i)\}_{i=1}^{n}$ with rotated bounding boxes
Require: Mask parameters: $p_{\text{obj}} = 4$, $\rho_{\text{obj}} = 0.75$, $p_{\text{bg}} = 16$, $\rho_{\text{bg}} = 0.5$
Require: Loss weights: α, β
for each image $x_i \in \mathbb{R}^{H\times W\times 3}$ **do**
  Initialize object mask $\Gamma_{\text{obj}} \leftarrow \mathbf{0}^{H\times W}$
  for each bounding box $b_j \in \mathcal{B}_i$ **do**
    Generate object region mask: $\mathcal{U}(b_j) \in \{0,1\}^{H\times W}$
    Apply fine-grained masking: $M_{\text{obj}} \sim \mathcal{M}(p_{\text{obj}}, \rho_{\text{obj}})$
    Update object mask: $\Gamma_{\text{obj}} \leftarrow \Gamma_{\text{obj}} \cup (\mathcal{U}(b_j) \odot M_{\text{obj}})$
  end for
  Generate background mask: $\Gamma_{\text{bg}} \sim \mathcal{M}(p_{\text{bg}}, \rho_{\text{bg}})$
  Compose final mask: Mask $\leftarrow \Gamma_{\text{obj}} \oplus \Gamma_{\text{bg}}$
  Apply masking: $x_{\text{masked}} \leftarrow x_i \odot \text{Mask}$
  $\mathbf{P} \leftarrow \phi(x_{\text{masked}}) + \mathbf{E}_{\text{pos}}$
  for $l = 1$ **to** L **do**
    $\mathbf{X}^{(l)} \leftarrow \text{TransformerBlock}(\mathbf{X}^{(l-1)})$ ▷ MSA + MLP
  end for
  $\hat{P} \leftarrow \mathbf{W}_d\mathbf{X}^{(L)} + \mathbf{b}_d$
  $w \leftarrow \log(s_{\text{object}}/s)$ ▷ Area-based weighting
  $L_{\text{Recon}} \leftarrow \sum_{\text{Mask}=1} w \cdot \text{SmoothL1}(x_i, \hat{P})$
  $L_{\text{TV}} \leftarrow \sum_{i,j}((\nabla_h x_i)^2 + (\nabla_v x_i)^2)^{\beta/2}$
  $L \leftarrow L_{\text{Recon}} + \alpha L_{\text{TV}}$
  Update model parameters via backpropagation
end for

Here, the Hadamard product ($\circ$) applies spatial weighting through a binary mask $\mathbb{1} \in \{0,1\}^{H\times W}$, while the logarithmic term $\log(s_{\text{object}}/s)$ dynamically scales loss contribution inversely with target size smaller objects receive exponentially higher weights to counteract their area disadvantage.

The second term L_{TV} implements edge-aware regularization to combat sensor noise and compression artifacts prevalent in aerial imagery

$$L_{\text{TV}} = \sum_{i,j}\left(\left(x_{i,j-1} - x_{i,j}\right)^2 + \left(x_{i+1,j} - x_{i,j}\right)^2\right)^{\frac{\beta}{2}}. \quad (19)$$

This generalized total variation norm ($\beta = 1$ recovers standard TV loss) penalizes high-frequency variations while preserving object boundaries through the exponent parameter β. The balancing coefficient α controls the tradeoff between detail preservation and noise suppression, optimized through systematic ablation studies as detailed in Section III.

G. Algorithm Pseudocode

The complete MSP-O pretraining procedure is formalized in Algorithm 1, and the MSP-O algorithm can be decomposed into four core phases: object and background masking, forward propagation, loss computation, and gradient backpropagation.

Key implementation details that are as follows.

1) Mask composition prioritizes object regions over background through logical OR operation with precedence.
2) Encoder processes all patches regardless of masking status using ViT architecture.
3) Area weighting w enhances small object reconstruction through logarithmic scaling.
4) TV loss gradients ∇_h, ∇_v compute horizontal/vertical intensity variations.

III. Experimental Results

A. Datasets

All experiments in this study were conducted on the oriented object detection datasets DIOR-R [28] and DOTA-v1.0 [29].

1) DIOR-R: It is a large-scale benchmark dataset for object detection in optical remote sensing images, featuring 20 target categories that include a range of geospatial objects such as airplanes, airports, and windmills. The dataset comprises 23 463 images with a uniform size of 800 × 800 pixels and spatial resolutions varying from 0.5 to 30 m. It is split into a training validation set with 11 725 images and a testing set with 11 738 images.

2) DOTA-v1.0: It is another extensive dataset for object detection in optical remote sensing images, consisting of 2806 images with varying dimensions from 800 to 4000 pixels in both width and height. It contains 188 282 individual instances across 15 classes. The dataset is divided into training, validation, and testing sets, with 1411, 458, and 937 images in each set, respectively.

B. Implementation Details

The experimental implementation integrates four ViT architectures (ViT-B [7], ViTAEv2-S [37], Swin-Tiny [36], and ViTAE-B [35]) through the MMpretrain framework [38], with all models initialized via MAE self-supervised pretraining [14]. To maintain geometric consistency across training phases, input images are standardized to 800 × 800 pixels using overlapping crops with 200-pixel increments, supplemented by random cropping and horizontal flipping for data augmentation. The optimization process employs cosine annealing scheduling with warm-up initialization [39], while detection fine-tuning utilizes oriented R-CNN [34] and RoI Transformer [40] implementations within the MMRotate framework [41]. All experiments are conducted on NVIDIA RTX 4090 and A6000 GPUs under identical hardware configurations, with detailed experimental settings (including hyperparameters and training protocols) thoroughly documented to ensure reproducibility.

C. Linear Probing of Pretraining

During the pretraining phase, the model was configured with 300 training epochs and an initial learning rate of 1×10^{-4}. The learning rate scheduling comprised two distinct phases: a linear decay regime for the first 40 epochs, followed by a cosine annealing strategy throughout the remaining 260 epochs to facilitate refined parameter optimization. To enhance computational efficiency during prediction head updates, we implemented a feature caching mechanism wherein backbone-derived features of test images were precomputed and archived in persistent storage, thereby saving the time consumed by repeated feature extraction.

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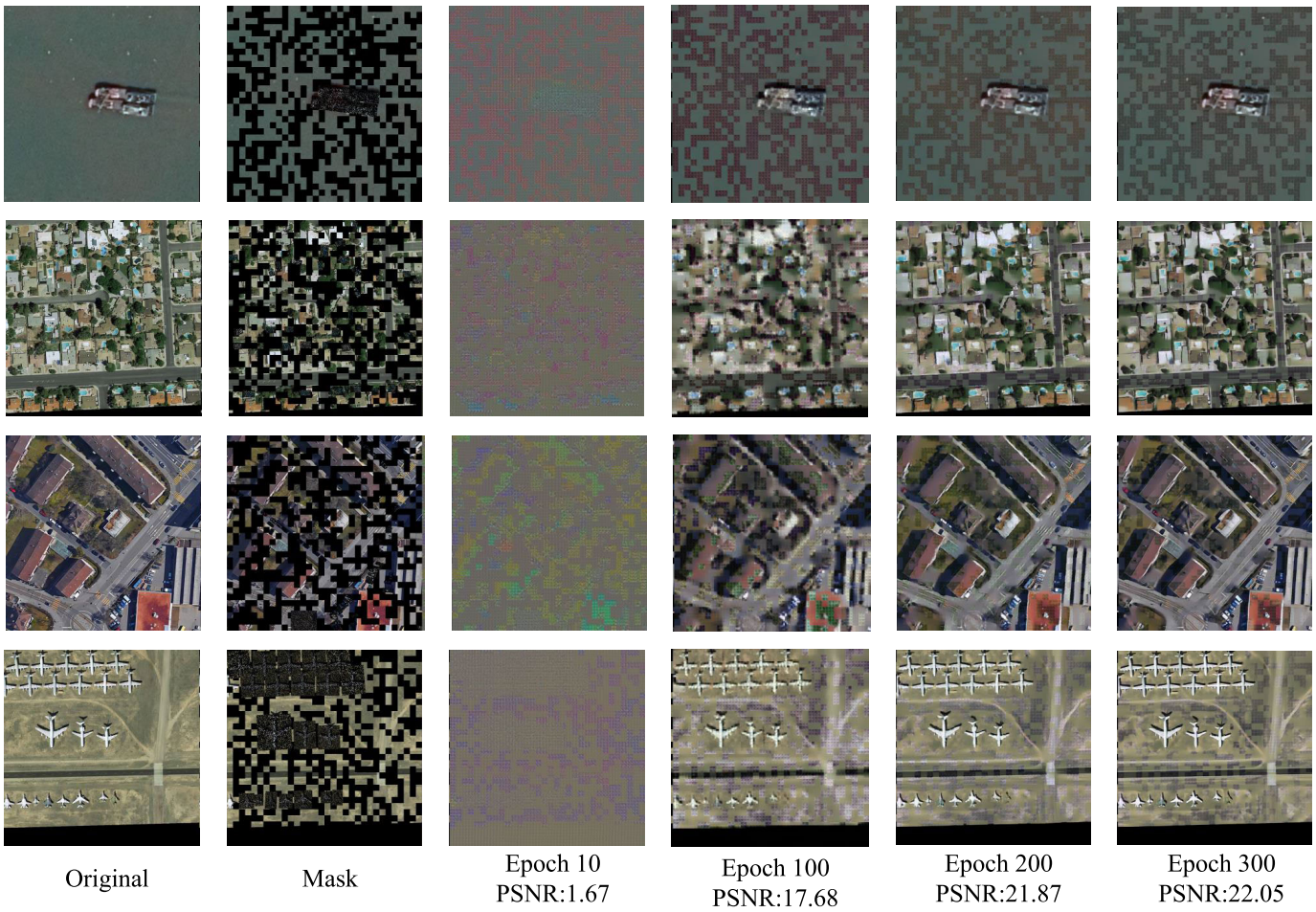


Fig. 4. Visualization of image restoration during the pretraining phase. The first column displays the original images and the second column shows the corresponding masked images. The subsequent four columns illustrate the progressively restored images at training epochs 10, 100, 200, and 300, respectively.

Pretraining Analysis: The reconstruction performance evolution of MSP-O pretraining is presented in Fig. 4. The quantitative evaluation of image restoration quality was conducted using the PSNR metric. Progressive improvement in denoising performance was observed with continued training. Notably, our empirical analysis revealed that enhanced noise suppression in image restoration does not exhibit a monotonic relationship with downstream task efficacy, suggesting potential over-regularization risks at advanced training stages.

We conducted oriented object detection experiments on the DOTA dataset using the oriented R-CNN framework. In this configuration, the backbone network parameters remained frozen while only the prediction head underwent supervised training. The model performance was quantitatively evaluated using the mean average precision (mAP) metric on the official testing set. Fig. 5 delineates three characteristic phases of the linear probing process.

1) *Phase 1 (Epochs 1–10):* The observed performance degradation temporally correlates with unstable reconstruction patterns in Fig. 4, indicating initialization mismatch between the prediction head's random weights and MSP-O pretrained features.
2) *Phase 2 (Epochs 10–100):* Rapid performance recovery occurred as the prediction head progressively aligned with the frozen backbone's feature representation space.

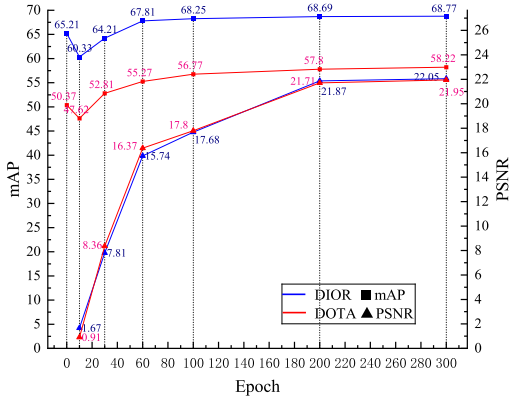


Fig. 5. Results of linear probing on DOTA and DIOR. The scale presented on the left side represents mAP. The performance of linear probing exhibits an initial decline followed by an upward trend, subsequently approaching a relatively stable state. The scale on the right side corresponds to PSNR. The quality of image restoration persists in a continuous growth pattern. This phenomenon implies that there is no definitive or deterministic correlation between the quality of image restoration and the precision of downstream tasks.

3) *Phase 3 (Epochs 100–300):* The improvement rate exhibited exponential decay characteristics, approaching an asymptotic convergence regime indicative of parameter optimization saturation.

The observed convergence dynamics motivated the adoption of 300 training epochs as the operational compromise between

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model performance stabilization and computational resource constraints. This configuration ensures sufficient parameter adaptation while avoiding diminishing returns from excessive training.

D. Fine-Tuning on Object Detection

To validate the effectiveness and generalizability of the proposed MSP-O framework, this article conducts a series of comparative experiments. Specifically, the DOTA and DIOR datasets are utilized for evaluation. For oriented object detection, oriented R-CNN [34] and RoI-Transformer [40] are employed as the detection models. The backbone networks used in the experiments include ViTAEv2-S [37], ViT-B [7], Swin-T [36], and ViTAE-B [35]. These backbone networks are initialized with advanced pretrained models, which consist of remote sensing supervised pretraining on the Million-AID and MAE self-supervised pretraining on the ImageNet. By conducting these comprehensive experiments, we aim to thoroughly assess the performance of the MSP-O framework across different configurations and demonstrate its superiority in the field of remote sensing object detection.

1) DOTA Dataset Evaluation: Quantitative evaluation results on the DOTA dataset under single-scale and multiscale detection protocols are summarized in Table II. For single-scale detection with 1024 × 1024 pixel crops and 200-pixel overlapping regions, the proposed MSP-O framework achieves consistent improvements across all configurations. The oriented R-CNN with ViT-B backbone attains 78.50 mAP (+1.44 mAP versus baseline), while the ViTAE-B variant reaches 78.68 mAP (+1.25 mAP improvement). The comparative analysis indicates stronger compatibility between MSP-O and oriented R-CNN compared with RoI Transformer counterparts, suggesting better synergy with oriented R-CNN's architectural principles. ViT backbones (particularly ViT-B) exhibit the most significant gains, demonstrating MSP-O's effectiveness in adapting standard ViT networks to multiscale remote sensing targets while narrowing performance gaps with hierarchical variants like ViTAE-B. Category-specific analysis reveals notable improvements for geometrically stable objects with distinctive features including airplanes, bridges, vehicles, and ships, confirming the framework's capability to extract environment-invariant features. These results collectively validate MSP-O's potential for oriented object detection in aerial imagery.

Under multiscale evaluation with 0.5×, 1.0×, and 1.5× input scaling, MSP-O consistently yields performance gains, albeit at a more moderate level. For oriented R-CNN, the mAP improvements range from +0.14 mAP (ViTAE-B: 80.64 versus 80.50 mAP) to +0.52 mAP (Swin-T: 79.07 versus 78.55 mAP). RoI Transformer achieves gains between +0.26 mAP (ViTAE-B: 80.51 versus 80.25 mAP) and +0.71 mAP (Swin-T: 79.54 versus 78.83 mAP). The relatively smaller improvement margins compared with single-scale scenarios are likely attributable to the inherent multiscale processing capabilities of baseline methods, which may be approaching performance saturation. The magnitudes of improvement across various categories are similar to those observed in single-scale scenarios, with ground targets of stable shapes and distinct features experiencing more pronounced enhancements. Nonetheless, MSP-O demonstrates complementary benefits across different resolution variations, thereby confirming its robustness to scale variance without the need for architectural specialization. These results collectively validate MSP-O's dual capability to enhance feature discriminability while maintaining compatibility with existing multiscale detection pipelines.

2) DIOR Dataset Evaluation: Quantitative evaluation results on the DIOR dataset are summarized in Table III, with cross-dataset validation conducted using native 800 × 800 resolution images validating the proposed methodology's enhanced generalization capabilities. Experimental results demonstrate consistent performance improvements across different backbone architectures: oriented R-CNN with Swin-T achieves 63.55 mAP (+0.36 mAP), ViT-B reaches 68.18 mAP (+1.78 mAP), and ViTAE-B obtains 65.11 mAP (+0.99). Parallel enhancements are observed in the RoI Transformer framework, where Swin-T yields 65.49 mAP (+0.24 mAP), ViT-B achieves 66.79 mAP (+0.69 mAP), and ViTAE-B attains 65.12 mAP (+1.00 mAP). Notably, per-category evaluation demonstrates maximum accuracy improvements for rotation-sensitive targets including ships (+2.45 mAP), airplane (+3.71 mAP), and ground track field (+3.88 mAP), confirming effective preservation of directional features. The Swin-T architecture shows significantly lower performance gains compared with ViT-B and ViTAE-B variants. This might be because its hierarchical structure has a limited capacity for modeling rotational invariance, unlike the global attention mechanisms of ViTs. This incomplete performance analysis requires further investigation through ablation studies on architectural components.

Qualitative comparisons visualized in Figs. 6 and 7 highlight the operational advantages of the proposed method. First, there is an enhanced detection capability for small and densely clustered targets, such as small ships and vehicles, with a significant reduction in the miss detection rate. In addition, the method shows a decreased number of false positive detections of background elements that exhibit geometric similarity to target objects, particularly for area targets like sports fields. Moreover, there is an improved precision in bounding box regression for irregularly shaped targets, such as airports. These visual improvements align with quantitative metrics, especially in challenging detection scenarios where traditional pretraining approaches struggle with rotational variance and scale diversity.

Comprehensive experimental evidence confirms that MSP-O effectively transfers structural knowledge through object-aware masking while maintaining detection robustness via spatial regularization. The consistent improvements across four distinct Transformer variants further demonstrate the architecture-agnostic nature of the approach, indicating its broad applicability in remote sensing detection pipelines.

E. Ablation Studies

The ablation analysis systematically examines two pivotal design parameters in MSP-O: pretraining data scale and masking strategy configuration. Evaluated on the DIOR dataset with

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TABLE II
RESULT OF DIFFERENT METHODS ON THE DOTA-V1.0 DATASET

Method	First pretraining[2]	Second pretraining	Backbone	PL[1] BD	BR GTF	SV LV	SH TC	BC ST	SBF RA	HA SP	HC	mAP (%)
Single-scale												
Oriented R-CNN	RSP	\	ViTAEv2-S	89.66 83.04	55.85 75.16	79.95 84.34	88.04 90.90	88.17 85.58	62.64 70.60	76.77 67.15	67.89	77.72
Oriented R-CNN	RSP	**MSP-O**	ViTAEv2-S	89.63 83.06	57.11 75.13	81.15 84.34	88.11 90.91	88.19 85.69	62.66 70.57	76.93 74.14	68.54	78.42 (+1.30)
Oriented R-CNN	MAE-I	\	Swin-T	85.31 83.88	54.62 73.73	80.57 82.89	84.61 85.82	83.72 66.02	56.84 69.27	73.93 76.76	70.37	75.22
Oriented R-CNN	MAE-I	**MSP-O**	Swin-T	87.40 84.92	56.69 75.75	80.65 82.90	86.62 87.83	84.77 67.10	60.93 69.28	74.17 76.81	70.43	76.42 (+1.20)
Oriented R-CNN	MAE-M	\	ViT-B	89.33 79.82	54.99 70.83	79.89 85.25	88.01 90.88	83.83 86.23	56.30 68.19	75.98 76.42	69.93	77.06
Oriented R-CNN	MAE-M	**MSP-O**	ViT-B	89.34 83.73	58.22 77.01	81.03 85.51	89.01 90.04	89.42 85.69	60.53 69.57	69.78 78.72	69.83	78.50 **(+1.44)**
Oriented R-CNN	MAE-M	\	ViTAE-B	89.40 80.02	55.25 70.90	80.84 85.48	88.20 90.87	83.56 86.39	56.54 69.87	77.11 79.63	67.47	77.43
Oriented R-CNN	MAE-M	**MSP-O**	ViTAE-B	91.44 83.53	55.47 75.11	81.30 84.97	89.21 90.00	89.72 85.90	67.40 70.73	68.33 76.19	70.93	**78.68** (+1.25)
RoI-Transformer	RSP	\	ViTAEv2-S	89.74 79.65	55.54 78.83	81.55 83.80	86.57 91.18	85.59 78.30	60.80 69.59	77.99 79.72	70.65	77.97
RoI-Transformer	RSP	**MSP-O**	ViTAEv2-S	89.86 79.83	58.77 79.07	81.95 83.89	88.84 91.22	85.71 80.42	61.02 69.61	78.35 79.22	71.01	78.58 (+0.61)
RoI-Transformer	MAE-I[2]	\	Swin-T	90.51 74.12	55.84 79.11	83.93 65.87	81.54 90.84	80.82 80.41	61.65 77.53	74.22 73.15	70.36	75.99
RoI-Transformer	MAE-I	**MSP-O**	Swin-T	90.82 76.16	58.43 70.74	86.72 68.35	85.88 90.92	80.90 82.12	61.55 70.34	77.11 76.55	74.07	76.71 (+0.72)
RoI-Transformer	MAE-M	\	ViT-B	86.02 77.98	58.69 75.18	77.92 83.04	86.86 86.02	78.75 88.67	67.01 69.88	78.20 75.51	69.91	77.31
RoI-Transformer	MAE-M	**MSP-O**	ViT-B	89.45 78.52	59.37 75.23	78.33 83.56	87.65 90.36	80.35 84.48	69.05 70.20	80.76 76.25	70.35	78.26 (+0.95)
RoI-Transformer	MAE-M	\	ViTAE-B	85.98 77.89	58.52 75.06	77.73 82.73	86.53 88.01	88.57 82.61	66.77 69.88	78.15 75.28	69.59	77.55
RoI-Transformer	MAE-M	**MSP-O**	ViTAE-B	85.91 77.91	58.73 76.10	77.91 83.00	86.79 90.96	88.69 85.64	66.98 71.82	78.12 75.44	70.84	78.32 (+0.77)
Multi-scale												
Oriented R-CNN	MAE-I	\	Swin-T	86.55 88.18	59.13 73.40	77.57 87.29	87.67 88.47	84.24 77.88	61.66 77.29	81.69 77.53	69.74	78.55
Oriented R-CNN	MAE-I	**MSP-O**	Swin-T	88.82 86.17	61.37 74.44	77.10 87.42	87.15 89.35	85.85 80.87	61.52 77.29	80.67 77.41	70.57	79.07 (+0.52)
Oriented R-CNN	MAE-M	\	ViT-B	89.46 84.32	61.81 79.74	79.66 85.54	88.47 90.85	82.19 86.58	62.04 72.48	84.63 80.72	78.43	80.46
Oriented R-CNN	MAE-M	**MSP-O**	ViT-B	89.47 84.51	61.43 79.76	81.62 86.52	88.44 90.87	87.72 86.52	62.44 72.67	84.62 77.37	79.41	**80.89** (+0.43)
Oriented R-CNN	MAE-M	\	ViTAE-B	87.47 84.90	60.97 79.23	80.62 85.08	88.48 90.84	85.66 87.32	62.11 70.27	83.82 82.03	78.67	80.50
Oriented R-CNN	MAE-M	**MSP-O**	ViTAE-B	91.22 88.11	61.37 81.05	81.10 84.91	87.27 90.84	81.21 85.07	65.47 69.91	81.22 81.82	79.09	80.64 (+0.14)
RoI-Transformer	MAE-I	\	Swin-T	86.41 88.89	58.13 80.09	78.47 83.09	86.89 86.23	79.54 89.07	61.29 70.20	78.28 80.98	74.87	78.83
RoI-Transformer	MAE-I	**MSP-O**	Swin-T	86.71 89.18	58.44 83.21	79.26 83.53	87.71 87.13	79.73 89.52	61.22 70.31	78.68 81.57	76.89	79.54 **(+0.71)**
RoI-Transformer	MAE-M	\	ViT-B	88.09 88.24	62.39 81.77	77.08 83.79	88.39 90.45	82.75 87.28	62.03 72.45	80.76 82.33	74.31	80.14
RoI-Transformer	MAE-M	**MSP-O**	ViT-B	88.92 88.07	63.71 82.66	81.69 84.02	89.99 90.90	80.47 86.37	62.18 71.38	80.04 82.40	75.03	80.52 (+0.38)
RoI-Transformer	MAE-M	\	ViTAE-B	90.42 88.98	57.79 82.87	79.16 84.43	87.23 91.09	79.63 87.13	63.48 76.13	80.94 82.91	71.56	80.25
RoI-Transformer	MAE-M	**MSP-O**	ViTAE-B	90.52 89.12	57.97 82.94	79.22 84.58	87.27 90.13	80.71 88.38	63.48 76.38	81.13 83.93	71.91	80.51 (+0.26)

[1] PL:plane. BD: baseball diamond. BR: bridge. GTF: ground track field. SV: small vehicle. LV: large vehicle. SH: ship. TC: tennis court. BC: baseball court. ST: storage tank. SBF: soccer ball field. RA: roundabout. HA: harbor. SP: Swimming pool. HC: helicopter.
[2] RSP: remote sensing supervised pretraining on the MillionAID. MAE-I: MAE unsupervised pretraining on the ImageNet. MAE-M: MAE unsupervised pretraining on the MillionAID.

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TABLE III
RESULT OF DIFFERENT METHODS ON THE DIOR-R DATASET

Method	First pretraining[2]	Second pretraining	Backbone	APL[1] APO	BF BC	BR CH	DAM ETS	ESA GF	GTF HA	OP SH	STA STO	TC TS	VE WM	mAP (%)
Oriented R-CNN	MAE-I	\	Swin-T	78.71 50.52	76.82 62.44	42.79 71.3	31.2 48.31	72.31 79.6	83.71 36.11	60.24 86.78	63.41 59.21	72.38 62.34	65.93 59.78	63.19
Oriented R-CNN	MAE-I	**MSP-O**	Swin-T	78.73 50.59	76.86 62.47	42.66 71.11	31.77 48.63	72.32 79.78	83.82 39.04	60.66 86.70	63.78 59.20	73.51 59.98	65.90 63.44	63.55 (+0.36)
Oriented R-CNN	MAE-M	\	ViT-B	80.04 41.86	80.79 81.39	44.83 78.35	35.12 67.67	84.85 75.44	80.80 37.65	59.33 81.15	78.70 62.87	89.83 56.17	49.87 65.36	66.40
Oriented R-CNN	MAE-M	**MSP-O**	ViT-B	83.75 43.84	81.31 89.46	43.99 80.44	36.95 74.30	71.98 81.37	84.68 39.61	81.37 83.60	86.11 53.64	80.20 43.88	54.94 68.26	**68.18 (+1.78)**
Oriented R-CNN	MAE-M	\	ViTAE-B	81.77 44.27	77.95 76.30	38.13 74.37	39.22 55.79	73.59 76.77	83.71 39.43	62.79 76.71	61.19 55.42	78.83 60.49	61.87 63.83	64.12
Oriented R-CNN	MAE-M	**MSP-O**	ViTAE-B	80.07 44.27	77.06 69.59	38.63 78.61	39.51 61.75	76.91 75.17	80.22 42.63	65.18 77.84	63.39 59.72	81.33 64.84	64.38 61.28	65.11(+0.99)
RoI-Transformer	MAE-I	\	Swin-T	80.19 42.34	78.32 79.80	39.39 80.64	35.82 65.04	77.20 69.33	80.96 46.71	58.64 81.14	61.76 70.42	85.74 55.20	50.31 66.12	65.25
RoI-Transformer	MAE-I	**MSP-O**	Swin-T	81.21 42.42	78.41 79.88	40.49 80.66	35.97 65.28	77.24 69.41	80.04 46.78	58.67 81.33	61.77 70.54	85.82 56.25	50.47 66.15	65.49 (+0.24)
RoI-Transformer	MAE-M	\	ViT-B	81.30 41.70	81.00 81.30	37.60 72.70	30.10 64.90	79.90 73.10	78.70 36.60	54.50 81.10	53.80 71.50	90.00 55.20	51.20 65.80	64.10
RoI-Transformer	MAE-M	**MSP-O**	ViT-B	81.31 41.77	81.02 83.62	39.97 72.73	30.42 65.03	80.23 73.45	81.01 38.95	54.61 82.51	54.10 71.91	90.43 55.35	51.47 65.98	64.79 (+0.69)
RoI-Transformer	MAE-M	\	ViTAE-B	81.77 44.27	77.95 76.3	38.13 74.37	39.22 55.79	73.59 76.77	83.71 39.43	62.79 76.71	61.19 55.42	78.83 60.49	61.87 63.83	64.12
RoI-Transformer	MAE-M	**MSP-O**	ViTAE-B	80.07 44.27	77.06 69.59	38.63 78.61	39.51 61.75	76.91 75.17	80.22 42.63	65.18 77.84	63.39 59.72	81.33 64.84	64.38 61.28	65.12 (+1.00)

[1] APL: airplane.APO: airport. BF: baseballfield. BC: basketballcourt. BR: bridge. CH: chimney. DAM: dam. ETS: expressway-toll-station. ESA: expressway-service-area. GF:golffield. GTF: ground track field. HA: harbor. OP: overpass. SH: ship. STA: stadium. STO: storage tank. TC:tenniscourt. TS: train station. VE: vehicle. WM: windmill.
[2] MAE-I: MAE unsupervised pretraining on the ImageNet. MAE-M: MAE unsupervised pretraining on the MillionAID.

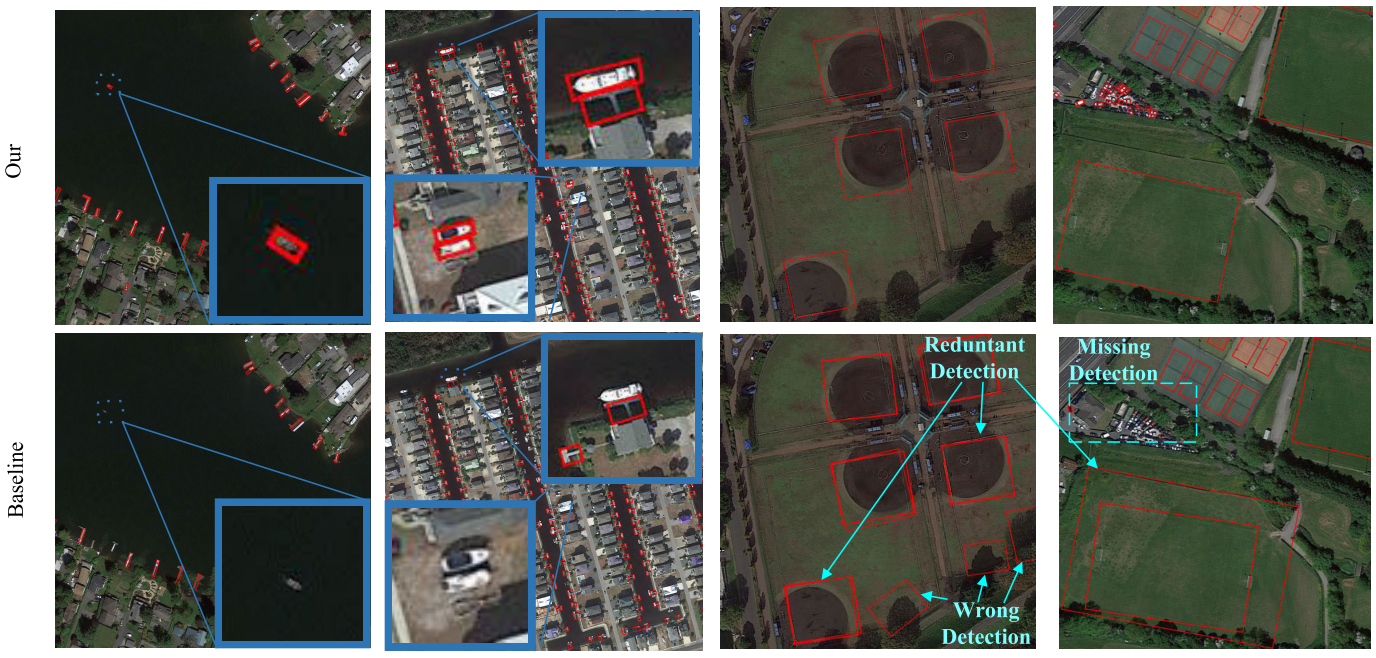


Fig. 6. Object detection results on the DOTA dataset, with cropped visualization images for clarity. The first columns present detection outcomes after MSP-O pretraining (our method), while the second columns display baseline results without such pretraining (baseline). Specifically, (a) and (b) emphasize that small objects (e.g., ships and cars) are frequently missed in the baseline, whereas (c) and (d) illustrate redundant and erroneous detections in the baseline, both of which are substantially diminished through MSP-O pretraining. This comparison vividly demonstrates that MSP-O pretraining effectively enhances detection accuracy and reduces errors by addressing critical limitations in small object recognition and false positive suppression.

ViT-B backbone under identical training protocols, the experiments first quantified the relationship between pretraining data quantity (20%–80% of full dataset) and downstream detection accuracy. As tabulated in Table IV, the mAP progressively increases from 60.21 to 66.89 mAP as the pretraining samples expand, establishing a direct correlation between data volume and feature representation quality a gain of 1.2 mAP per 10% data increment. The proposed copy-embedding strategy demonstrates consistent performance enhancements across all data regimes, most significantly improving small-data

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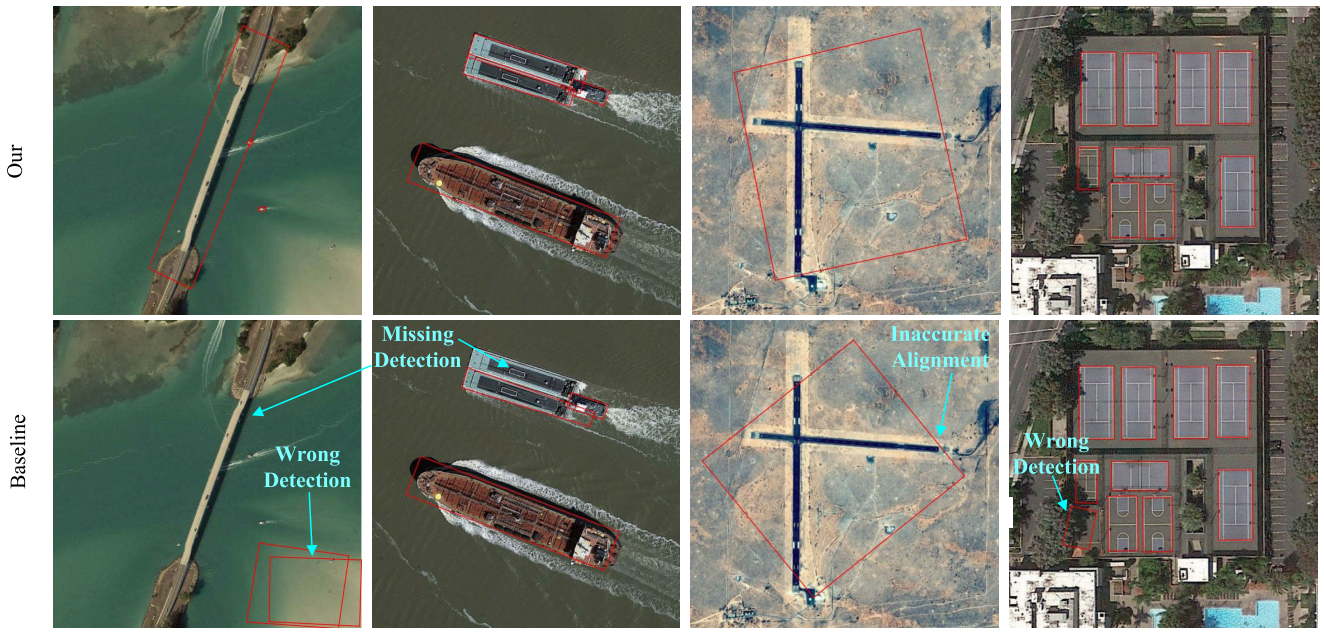


Fig. 7. Object detection results on the DOTA dataset, where the visualization images have been cropped. The first column presents the detection results after MSP-O pretraining (our method) and the second column shows the detection results without MSP-O pretraining (baseline). Specifically, (a) and (b) evident that long and narrow objects, such as bridges and ships, are often missed in the baseline results. Regarding (c), angle localization of irregular targets, like airports, is more accurate. (d) False detections of area targets, such as basketball courts, are decreased.

TABLE IV
IMPACT OF THE NUMBER OF IMAGES AND COPY-EMBEDDING STRATEGY ON DOWNSTREAM TASKS

Pretrained Images	Copy-Embedding Strategy	mAP (%)
2k (20%)	×	60.21
	✓	64.77 (↑4.56)
4K (40%)	×	64.15
	✓	67.05 (↑2.90)
5K (50%)	×	65.76
	✓	67.87 (↑2.11)
6K (60%)	×	67.02
	✓	67.88 (↑0.86)
8K (80%)	×	66.89
	✓	67.92 (↑1.03)

TABLE V
MASK RATE CONFIGURATION ANALYSIS FOR OBJECT AND BACKGROUND REGIONS

Background Mask (%)	Object Mask (%)	mAP (%)
35	60	65.24
	75	**65.43**
	90	65.05
50	60	66.23
	75	**67.87**
	90	67.55
35	75	67.05
50		**67.87**
75		67.55

[1] Bold values indicate optimal configurations per section

scenarios: with only 20% pretraining data, mAP reaches 64.77 mAP (+4.56 mAP versus baseline 60.21 mAP), surpassing the 40% baseline performance (64.15 mAP). This demonstrates the capacity of the strategy to effectively mitigate data scarcity through feature reuse, which is particularly valuable for remote sensing applications with limited annotated samples.

Further investigations evaluate masking strategy impacts through controlled parameter variations, revealing critical insights into information preservation-learning difficulty trade-offs. With background masking fixed at 50%, increasing object masking rates from 60% to 90% reveals a nonlinear response mAP peaks at 75% masking (67.87 mAP) before declining to 67.55 mAP at 90%, suggesting structural information degradation under excessive object occlusion outweighs increased learning challenges. Parallel experiments with background masking fixed at 35% exhibit similar parabolic trends, confirming the robustness of this relationship across configurations. Conversely, maintaining 75% object masking while adjusting background rates from 30% to 70% yields comparable nonmonotonic behavior, achieving optimal performance (67.87 mAP) at 50% background masking. These empirical patterns corroborate masked autoencoder principles, where partial masking achieves an optimal equilibrium between context prediction complexity and feature preservation fidelity. Despite computational constraints, Table V empirically validates our adaptive masking principle: the hybrid configuration (75% object + 50% background masking) yields 1.64 mAP gains over conventional uniform masking baselines (50% object + 60% background). The findings confirm the necessity of domain-specific mask ratio tuning for detector optimization, particularly in remote sensing imagery with distinct target-background distributions characterized by scene-dependent saliency variance.

F. Transfer Studies

To systematically evaluate the cross-dataset generalization capabilities of MSP-O, we established a comprehensive

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TABLE VI
CROSS-DATASET TRANSFER LEARNING PERFORMANCE WITH VIT-BASED BACKBONE

Source Dataset	Target Dataset	mAP (%)
–	DOTA	77.05
DIOR	DOTA	77.55 (↑0.50)
DOTA	DOTA	78.56 (↑1.51)
–	DIOR	66.65
DOTA	DIOR	67.48 (↑0.83)
DIOR	DIOR	68.18 (↑1.53)

[1] Training from MAE pretraining weight
[2] ↑ indicates improvement over baseline

TABLE VII
CROSS-DATASET TRANSFER PERFORMANCE COMPARISON BETWEEN MSP-O PRETRAINING AND DIRECT FINE-TUNING APPROACHES USING ORIENTED R-CNN WITH VIT-BASED BACKBONE

Source Dataset	Target Dataset	Transfer Method	mAP (%)
–	DOTA	–	77.05
DIOR	DOTA	Direct Fine-tuning	77.15 (↑0.10)
DIOR	DOTA	MSP-O Pretraining	77.55 (↑0.50)
–	DIOR	–	66.65
DOTA	DIOR	Direct Fine-tuning	66.65 (↑0.00)
DOTA	DIOR	MSP-O Pretraining	67.48 (↑0.83)

[1] Training from MAE pretraining weight
[2] ↑ indicates improvement over baseline

cross-dataset transfer learning framework involving bidirectional domain adaptation experiments. Our experimental design incorporates two transfer paradigms: DOTA→DIOR pretraining followed by domain-adaptive fine-tuning, and its reciprocal configuration DIOR→DOTA. As detailed in Table VI, cross-dataset pretraining consistently outperforms non-pretrained baselines, achieving mAP improvements of 0.50 mAP (DOTA→DIOR) and 0.83 mAP (DIOR→DOTA). Notably, while these gains validate MSP-O's cross-dataset transfer efficacy, they remain comparatively modest relative to same-domain pretraining scenarios, with performance differentials of 1.01 mAP (DOTA→DOTA versus DIOR→DOTA) and 0.70 mAP (DIOR→DIOR versus DOTA→DIOR).

The experimental results reveal two key insights. First, the asymmetric performance characteristics in bidirectional transfers (0.50 versus 0.83 mAP improvements) suggest domain-specific adaptation challenges potentially arising from dataset heterogeneity in object scales or sensor characteristics. Second, the persistent performance gap between cross-dataset and intradataset pretraining underscores the inherent complexity of knowledge transfer across aerial imaging domains. These findings collectively demonstrate the capability of MSP-O in facilitating interdomain knowledge transfer while highlighting promising research directions for addressing domain shifts in remote sensing applications.

Furthermore, a controlled comparative study was conducted to isolate the contribution of pretraining MSP-O versus standard detection task transfer. The experimental pipeline involved initial fine-tuning of the object detector on DOTA, subsequently repurposing the learned parameters as initialization for DIOR fine-tuning. Quantitative results in Table VII demonstrate the statistically significant superiority of MSP-O-based transfer over conventional task-specific transfer paradigms. The enhanced accuracy metrics indicate that the reconstruction objective of MSP-O facilitates the learning of more transferable feature representations, which capture both domain-invariant patterns and dataset-specific characteristics. This demonstrates the superior representation learning capability of our proposed pretraining strategy.

IV. CONCLUSION

This study introduces MSP-O, an innovative two-stage pretraining method for remote sensing image object detection. MSP-O inserts a pixel-level restoration phase between traditional pretraining and model fine-tuning to sharpen the model's focus specifically on target areas. This significantly boosts detection accuracy in complex backgrounds and across diverse categories. MSP-O outperforms existing methods on the DOTA and DIOR datasets, underscoring the value of effective pretraining in remote sensing, especially with limited data. Our work expands pretraining applications and lays a solid foundation for future research in remote sensing object detection.

Future work will extend the research to include multimodal datasets, such as infrared, hyperspectral, and synthetic aperture radar. The current research focuses mainly on the Transformer architecture, and future studies will explore the potential of CNN architectures to further assess their applicability and impact on the field of remote sensing. In addition, while this article primarily focuses on object detection tasks, we believe that the proposed method is also applicable to other remote sensing applications, such as scene segmentation.

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